

Work Note (v2) — Incorporating Published Literature Data into the Lab OCM Yield Model

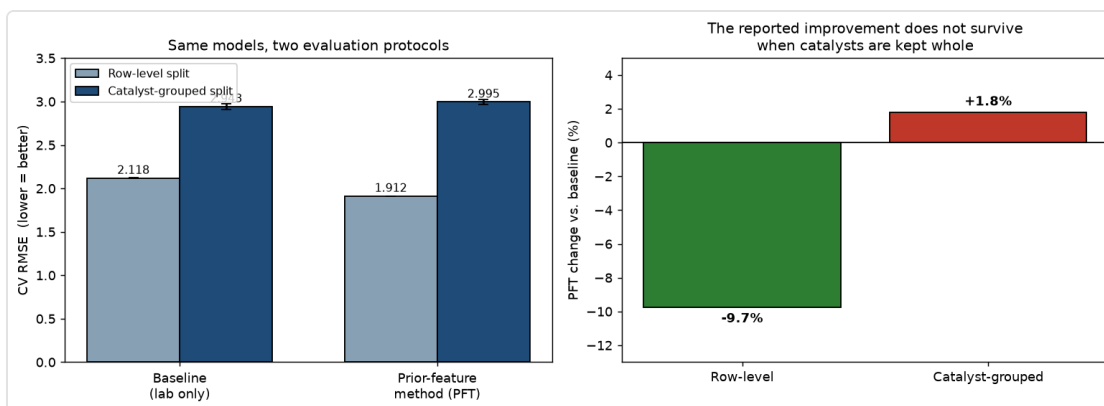
To: Prof. Taniike **Topic:** A domain-adaptation study for C₂-yield prediction in Oxidative Coupling of Methane (OCM) **Supersedes:** version 1 of this note. **Companion code:** `ocm_methodology.ipynb`, `taniike_validation.py`, `phase3_lit_prior.py`, `phase4_family_diagnosis.py`, `phase5_target_audit.py`, `phase6_our_experiments.py`, `phase6_candidates.py`

What changed since version 1

Version 1 reported that a two-stage prior-feature method ("PFT") improved C₂-yield prediction by about 10 % over a lab-only baseline. **Following the stricter validation Prof. Taniike proposed, that improvement does not survive, and we withdraw the claim.**

Claim in v1	Status in v2
PFT improves CV RMSE by ≈ 10 %	Withdrawn. The gain was catalyst-identity leakage; under catalyst-grouped CV, PFT is 1.8 % worse than baseline
Literature data measurably helps the lab model	Not demonstrated. Four honest designs, all $\leq$ the composition-only control
Quantile normalisation is a necessary component	Unnecessary. Quantile-normalised $\approx$ raw $\approx$ rank priors (0.001–0.005 RMSE apart)
—	New: the composition-only model screens <i>unseen</i> catalysts usefully ($\rho = 0.761$, enrichment 3.04–4.89 $\times$)
—	New: the Ba-family failure is mechanistically explained, and yields a data budget for new chemistry

The cause was specific and is now understood: our Stage-1 expert was trained on literature data *together with the lab training rows*, so under a random row split it had seen the very catalysts it was later asked to help predict. What follows is the corrected study.



1. Objective

We predict C_2 yield from catalyst composition using the lab's own data (89,074 measurements, 2025, impregnation route). A body of published literature data (3,852 measurements, 1982–2019) is also available. The question is whether the literature can improve the lab model without degrading it.

Two distributional differences make this a **domain-adaptation** problem: a **label shift** (literature mean 8.670 % vs. lab 5.245 %, a +3.425 pp gap reflecting publication practice) and a **covariate shift** (the two collections emphasise different chemistry).

2. Evaluation methodology

Catalyst-grouped cross-validation is now our default protocol. All measurements of a given catalyst are assigned to exactly one fold, so every reported number answers: *how well do we predict a catalyst nobody has made yet?* Anything fitted on data — scaler, domain classifier, both model stages — sees training-fold data only.

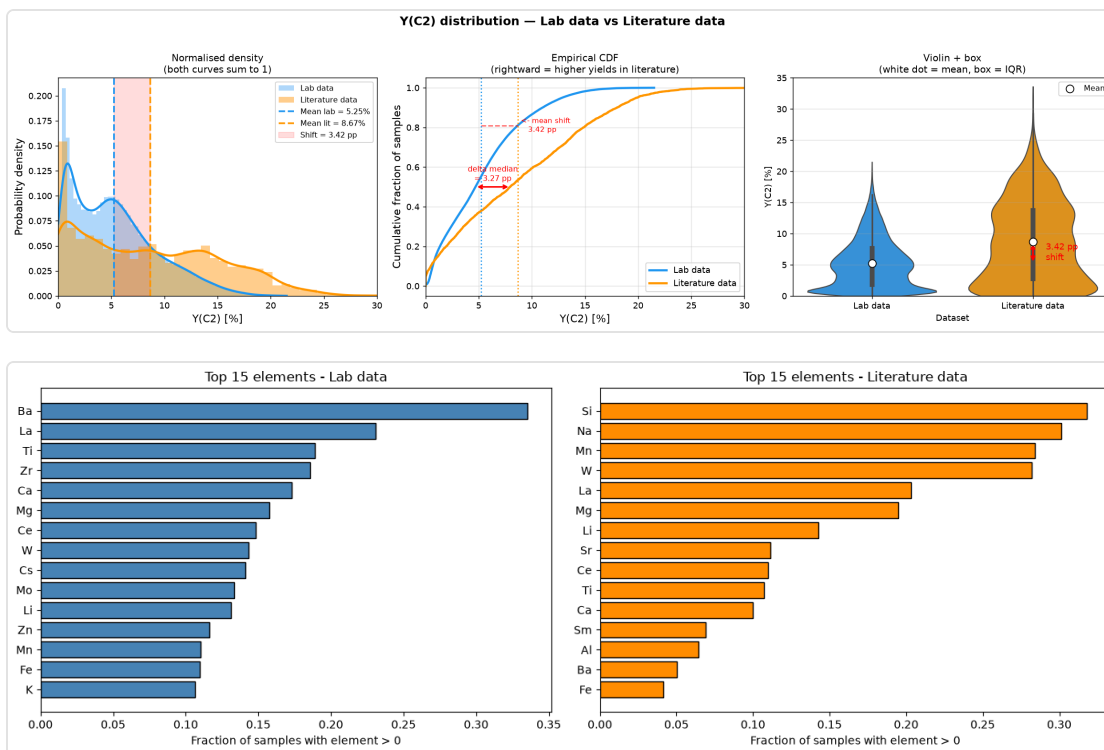
Why the previous protocol was inadequate — a quantitative argument. The 89,074 measurements comprise only **917 distinct catalysts** at 5 temperatures, giving ~4,400 unique input vectors each measured ~20 times. Because roughly 27 reaction-condition variables are not recorded as features, those repeats are *identical inputs with differing yields*. That within-group variance is **18.2 % of total yield variance**, so the best attainable row-level RMSE is **1.680**.

Version 1 reported 1.912 — only 0.23 above that floor. In hindsight this should itself have prompted suspicion: a model cannot approach the noise ceiling of data it should not be able to memorise. Point-wise RMSE is therefore not merely less relevant than catalyst-level metrics here; it is close to uninformative.

Primary metrics are consequently catalyst-level, as proposed: Spearman correlation between predicted and observed *maximum* yield, and enrichment of true high performers among top-ranked predictions.

3. Baseline and the two datasets

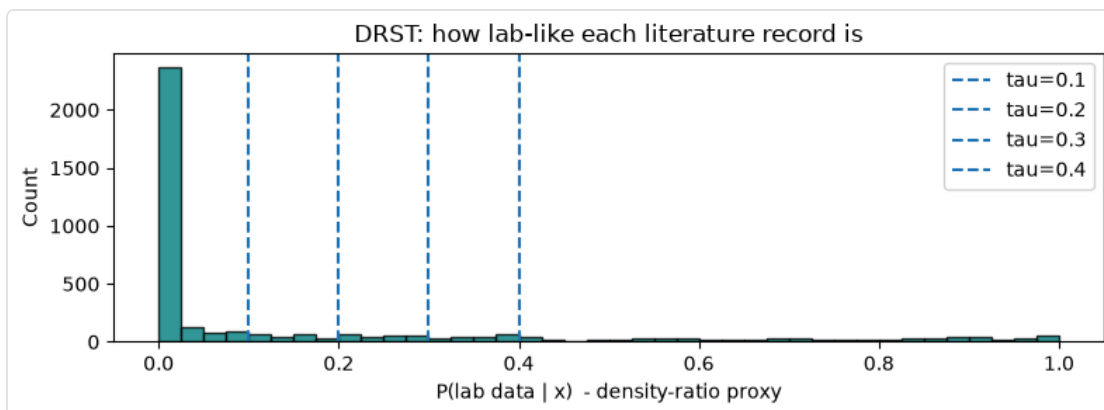
A LightGBM regressor trained on lab data only. Under the grouped protocol its CV RMSE is **2.943** (row-level: 2.118 — the difference is the leakage, not a change of model).

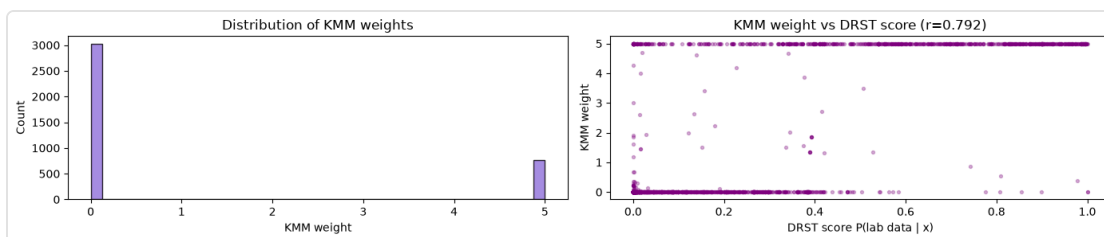


4. Approaches evaluated

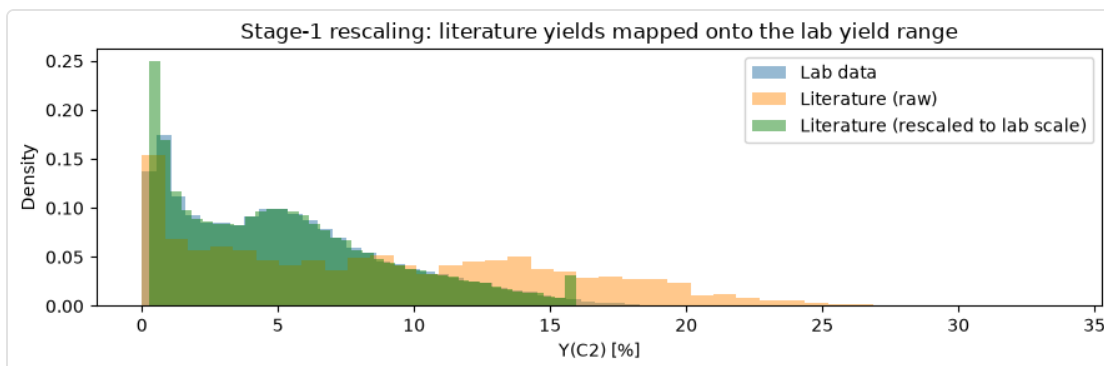
Direct merge. Pool all literature records with the lab data. The label shift enters the training target and accuracy is not improved.

Selective merge. Make the literature more lab-like before merging — either a hard filter (DRST: a logistic-regression domain classifier scoring each record by $P(\text{lab} \mid \text{features})$, retaining 20.3 % at $\tau = 0.30$) or continuous importance weights (KMM). Both still place literature yields in the training objective, so the label shift is unresolved.





Prior-feature method (PFT). Let the literature influence the final model only through a *predicted value used as an input feature*, never as a training label: Stage 1 trains an expert on literature data whose yields are rank-rescaled onto the lab scale; Stage 2 trains on lab data with that prediction as one extra feature, using lab labels only.



5. What the stricter validation showed

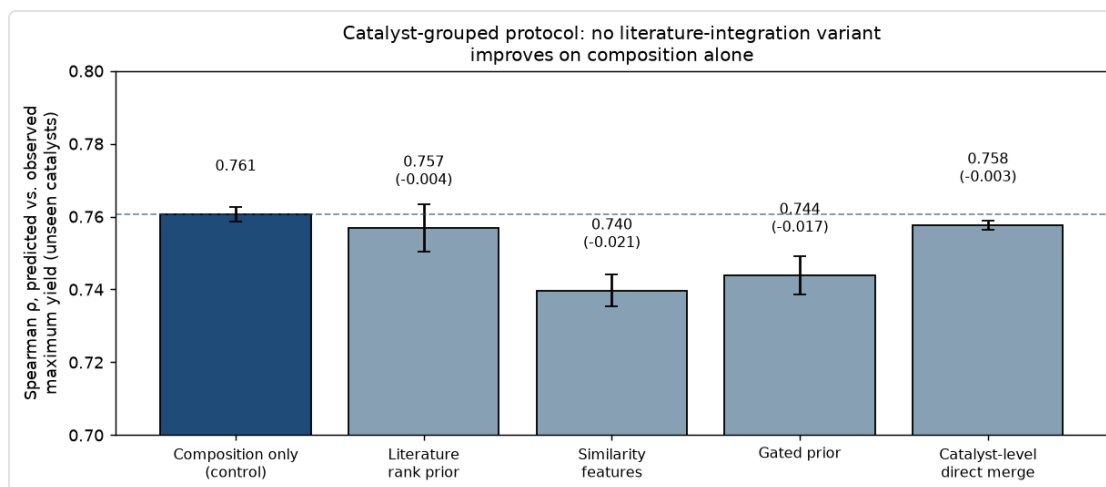
The improvement was leakage. Under catalyst-grouped CV: baseline **2.943**, PFT **2.995** (+1.8 %). Three independent checks establish the mechanism:

1. Training Stage 1 on literature *alone* reduces the row-level gain from -9.7 % to **-2.6 %**
2. Under grouped CV the joint variant (**2.982**) is *worse* than literature-only (**2.938**)
3. Literature-only under grouped CV (2.938) is indistinguishable from baseline (2.943)

Your quantile-normalisation hypothesis was correct, and the effect is essentially exact.

Quantile-normalised, raw-yield and pure-rank priors differ by only 0.001–0.005 RMSE under both protocols. The Stage-2 trees consume only the prior's ordering, so the normalisation step can be removed with no loss.

With the leakage channel closed, we retested literature integration properly — a literature-only rank prior, similarity-to-literature features, a gated prior combining both, and a catalyst-level direct merge. Success criteria were fixed before running. **None improved on composition alone.**



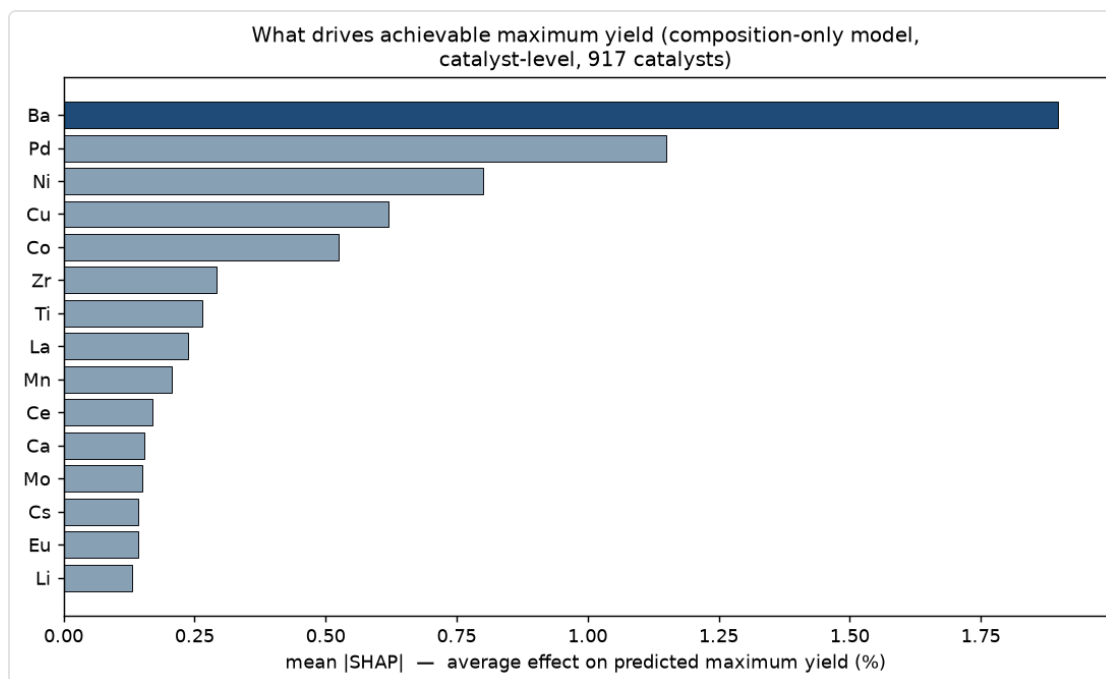
We also tested a hypothesis of our own — that the prior might help specifically where our own coverage is thin, suggested by one family (Zr) showing a consistent gain. Across **all 28 element families with ≥ 50 catalysts** the mean effect was **-0.0025**, 14/28 positive, and the strongest correlation with any coverage measure was $|\rho| = 0.276$ against a pre-registered threshold of 0.5. The Zr result was selection from noise, and we discarded it.

6. What the model can do: screening unseen catalysts

Rebuilt at catalyst level — composition $\rightarrow$ maximum yield, 917 training examples, no temperature — the model matches the full 89,074-row model on ranking while training $\sim 100\times$ faster. On genuinely unseen catalysts:

Metric	Value	95 % CI
Spearman ρ (predicted vs. observed max yield)	0.761	0.725 – 0.785
Enrichment of true top-decile among top-decile predicted	4.28×	3.04 – 4.89×
Precision@20 (of 20 nominated, fraction truly top-decile)	0.44	0.15 – 0.65

We quote intervals rather than point estimates: with ~ 18 catalysts in a test decile, these quantities are considerably less precise than a single number suggests.



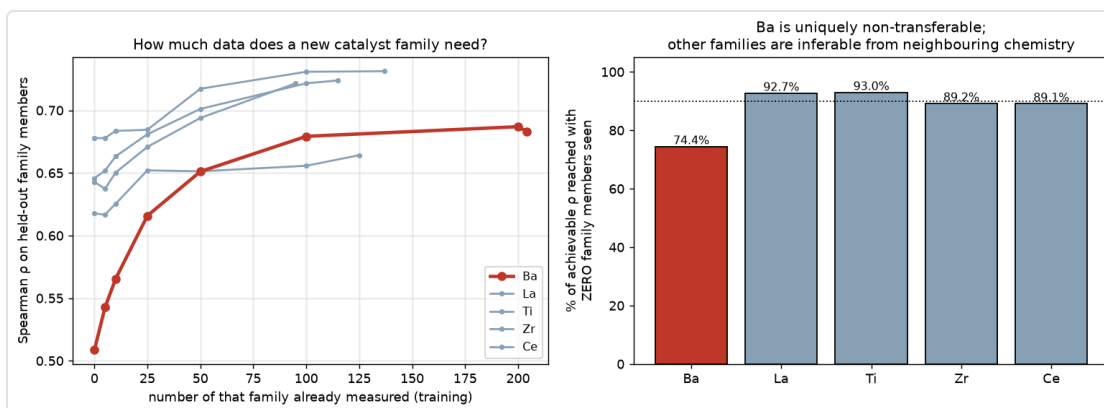
7. Why the Ba family fails, and how much data a new family needs

Family holdouts behave reasonably for La, Ti, Zr and Ce (ρ 0.62–0.68) but poorly for Ba (0.526). The cause is quantifiable: Ba catalysts average **13.76 %** maximum yield against **8.95 %** for the rest, and **78 % of the lab's top decile contains Ba**. Removing them removes the high-yield regime itself.

The mechanism is provable rather than inferred. With no Ba catalysts in training, the Ba column is constant, so no tree splits on it — and retraining with that column **deleted entirely** produces **bit-identical predictions**. The model prices Ba catalysts as though Ba were absent, underpredicting the best of them by **9.8 yield points**. Ten random pseudo-families of equal size score 0.752, so this is chemistry and label coverage, not sample size.

That raised a question we found more useful than *whether* it fails: **how many labelled members of a family are needed before the model can price it?**

Family	ρ , none seen	ρ , fully seen	% of ceiling at zero
Ba	0.509	0.683	74.4 %
La	0.678	0.731	92.7 %
Ti	0.618	0.664	93.0 %
Zr	0.646	0.724	89.2 %
Ce	0.643	0.722	89.1 %



Ba is uniquely non-transferable: the other families are already at 89–93 % of their ceiling having never seen a member, because neighbouring chemistry carries the information. Ba's curve is still rising at $n = 200$ while the others saturate near $n \approx 50$. **As a practical budget: a chemically distinctive new promoter family requires roughly 50–100 labelled catalysts before it can be modelled; a family resembling existing chemistry requires 10–25.**

8. A candidate list for prospective validation

We enumerated **26,414 unseen candidates** in the laboratory's own design grammar — impregnation, one support at ~90 % with 2–3 promoters at ~3.33 %, drawn from the supports and promoters already in use — and scored them with a 10-seed ensemble. No literature prior is used.

Every candidate carries a coverage flag, and the safeguard is verified rather than assumed: for an element absent from our data (Ag), predictions computed with and without its column differ by exactly zero — confirming such candidates are unpriceable and must be flagged rather than ranked.

The highest-ranked candidate is **Ba(90) + Mo(3.33) + Zn(3.33) + Fe(3.33)**, predicted 18.79 % $\pm$ 0.09.

Two caveats. Absolute predictions compress at the extreme — our ensemble maximum is 18.79 % while observed training yields reach 21.50 % — so the **ranking** is the deliverable, not the predicted value. And the top 20 are chemically monotonous: all contain Ba, most contain Mo. A diversity constraint did not meaningfully change this, because the model's preference for Ba is genuine. We therefore also provide the best candidate per support, with its cost in predicted yield:

Support	Ba	Ti	La	Ca	Mg	Si	Al	Zr	Ce
Best predicted max yield (%)	18.79	16.49	15.82	15.54	14.66	13.83	13.63	13.02	12.93

Our recommended campaign (17 catalysts, `campaign_shortlist.csv`). Rather than submit twenty near-identical catalysts, we suggest splitting the budget:

- **Tier A — 12 catalysts, the model's optimum** (predicted 18.33–18.79 %). This is where the expected hits are. From the retrospective grouped-CV precision@20, we would expect roughly

2–8 of the 12 to be genuine top-decile performers (95 % CI 0.15–0.65 — deliberately a wide interval).

- **Tier B — 5 catalysts, one per alternative support** (Ti 16.49, La 15.82, Ca 15.54, Mg 14.66, Si 13.83). These cost predicted yield and we do not expect them to win. They are an *information* purchase: 78 % of the lab's existing top decile already contains Ba, so the model's strong Ba preference may partly reflect that coverage rather than chemistry. If several Tier-B catalysts outperform their predictions, that tells us something the Tier-A catalysts cannot.

We would of course defer to your judgement on synthesis feasibility, and are happy to reweight the split.

9. Relation to prior work

The mechanism of PFT — using a model's prediction as an input feature — is **stacked generalisation** (Wolpert, *Neural Networks* 5, 1992, 241–259) applied across distributions. "Prior Feature Transfer" was our internal shorthand, not a standard term, and we note that "feature transfer" usually denotes *representation* transfer, which this is not.

Our corrected understanding of when such a method can help. Because the prior $p = f(x)$ is a deterministic function of features the final model already has, $I(y; x, f(x)) = I(y; x)$ — it cannot add information. It can only supply an inductive bias, or exploit source data covering regions the target data does not. Our results indicate neither applies here at the scale that matters, for three reasons: ~78.5 % of the literature is out-of-distribution relative to the lab; quantile normalisation aligns marginal but not conditional distributions; and literature yields are measured under each paper's own conditions, whereas our target is the maximum over a standardised battery — semantically different quantities. **The honest scope statement is that this family of methods is a local-coverage tool, not a global-information tool.**

Related approaches remain distinct: Δ -machine-learning (Ramakrishnan et al., *JCTC* 11, 2015, 2087) adds a source estimate as an additive baseline, inheriting its scale; importance weighting (Huang et al., NIPS 2006; Sugiyama et al., *JMLR* 8, 2007) corrects which records are used but keeps source labels in the objective — our DRST and KMM baselines are instances; label-shift correction (Lipton et al., ICML 2018) reweights to correct the label distribution.

10. Limitations and next steps

- **The literature contribution is not demonstrated.** Four designs plus a 28-family follow-up all returned null. We report this rather than continue searching for a variant that scores well.
- **Novel promoter families cannot be priced.** This is structural, not a modelling deficiency — but §7 quantifies the data required to remove the limitation.
- **The target carries a measurement-effort confound.** 47 catalysts have fewer than 20 measurements and systematically low maxima; part is a pure sampling effect, part appears to be deliberate early stopping. Excluding them changes our headline metric by 0.002, so nothing hinges on it — but we would welcome confirmation of which it is.

- **The unrecorded reaction conditions are the largest single opportunity.** If those ~27 variables exist in a retrievable form, the 18.2 % irreducible variance becomes partly learnable and condition-level modelling becomes meaningful.
- **Cross-preparation transfer is now measured, and it is where literature data finally helps.** Predicting *impregnation* literature (different source, same preparation) gives $p = 0.398$; predicting *non-impregnation* literature gives $p = 0.238$ — so changing preparation costs a further 0.160. At that range the model is not usable for selection out of preparation: its top-decile picks average 11.35 % true yield against a population mean of 10.33 %, while the genuine top decile averages 21.88 % (enrichment 0.42 $\times$, i.e. no better than random). **But adding impregnation literature to training raises p from 0.238 to 0.388 (+0.150, 5/5 seeds)** — the first setting in this study where literature data measurably helps, and consistent with the scope statement in §9: the lab has *no* coverage of non-impregnation chemistry, so the literature supplies genuinely new information there. Two caveats: absolute performance remains poor, and plain merging outperforms the prior-feature construction (0.388 vs 0.318). Details in `phase7_prep_ood.py`; this supersedes the earlier row-level OOD numbers.
- **Prospective validation** is the natural next step, and we would pre-register the expected hit rate (precision@20, CI 0.15–0.65) before any results arrive.

All numbers in this note are stored outputs of the scripts named at the head of the document; each figure is generated from the corresponding experiment's JSON so that figures cannot drift from the experiments that produced them.